PRABHAT DUBEY

Senior Software Engineer | AI Infrastructure | Distributed Systems | LLM/GenAI | Data Protection
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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of experience building high-throughput distributed systems, AI/ML infrastructure, and cloud-native platforms at scale. Architected enterprise-grade solutions processing petabyte-scale data across AWS and Azure ecosystems. Currently building LLM-powered intelligent systems with RAG pipelines and AI agents for enterprise automation. Inventor on 14 granted US patents in cloud storage and virtualization. Proven track record of shipping zero-downtime features, driving cross-functional engineering teams, and integrating AI capabilities into production systems. Currently pursuing M.Tech in AI/ML from BITS Pilani.

TECHNICAL SKILLS

Languages: C++, Java, Python, C#, Go, TypeScript
AI / ML / LLM: PyTorch, LangChain, HuggingFace, RAG, Fine-tuning, Prompt Engineering
GenAI Stack: OpenAI API, Pinecone, ChromaDB, AI Agents, Embeddings
Frameworks: Spring Boot, .NET Core, Angular, Node.js, FastAPI
Cloud & Infra: AWS (S3, SageMaker, Lambda), Azure (Blob, OpenAI), Kubernetes, Docker
Databases: PostgreSQL, MongoDB, Redis, SQLite, Vector Databases
System Design: Distributed Systems, Microservices, ML Pipelines, Model Serving
MLOps & DevOps: MLflow, Jenkins, Git, CI/CD, Weights & Biases, REST/gRPC

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Dell Technologies | Bengaluru, India | Apr 2022 – Present

- Architected SQL Server per-file database relocation feature end-to-end (UI to backend across 6 repositories), reducing the original epic delivery estimate by 70% and enabling zero-downtime restore operations.
- Designed and built Native Edge Plugin Service using Go and Java microservices to automate onboarding and discovery of 1,000+ edge devices, improving setup efficiency by 30% and cutting provisioning time from hours to minutes.
- Engineered vProxy discovery daemon for automated VM discovery in Hyper-V and Native Edge environments, reducing manual intervention by 40% and eliminating configuration drift across distributed clusters.
- Led the Hyper-V Backup & Restore module integrating C++, .NET, and Java services for Dell PowerProtect Data Manager (PPDM), achieving 99.9% restore reliability across 50+ enterprise customers.
- Built GHVDM microservice layer (Spring Boot) bridging PPDM with Hyper-V Agent, processing 10K+ metadata operations/day with sub-second latency for data flow and metadata management.
- Developed AI-powered log analysis pipeline using LLM-based classification and RAG to auto-diagnose backup failures from unstructured logs, reducing mean-time-to-resolution by 45% across support workflows.
- Resolved 15+ critical P0/P1 customer escalations under SLA, mentored 5 engineers on system design, debugging, and performance tuning best practices.

Software Engineer II, Dell Technologies | Bengaluru, India | Dec 2019 – Mar 2022

- Developed Object Store backup/restore SDK for Azure Blob and AWS S3 using C++ and Node.js, improving cloud data transfer throughput by 35% through parallel chunked uploads and connection pooling.
- Architected Active Directory Granular Level Restore (AD GLR) system with iSCSI-based live mount, reducing recovery turnaround from hours to minutes for enterprise AD environments.
- Served as Scrum Lead for AD GLR epic; managed sprint planning, backlog refinement, and delivery tracking for a team of 8 across 4 sprints.
- Optimized backup verification pipeline using SQLite and VSS integration, reducing recovery time by 25% through query optimization and batch processing.
- Named inventor on 14 granted US patents covering cloud storage optimization, Office 365 backup, and virtualization recovery techniques.

Software Engineer, Micro Focus (now OpenText) | Bengaluru, India | Aug 2016 – Nov 2019

- Built core modules for Directory and Resource Administrator (DRA) product using C++/.NET backend and Angular frontend, improving feature delivery velocity by 20%.
- Developed enterprise desktop applications using WCF and WinForms for administrative workflows, service communication, and configuration management within the DRA ecosystem.
- Designed WCF-based service layers enabling secure, encrypted communication between management consoles and backend services across distributed Active Directory domains.
- Created standalone Health Check Utility (WinForms) for product validation and diagnostics; recognized at internal hackathon for innovation and adopted org-wide.

KEY PROJECTS

- **Intelligent Log Diagnostics (RAG + LLM):** Built a Retrieval-Augmented Generation pipeline using LangChain, ChromaDB, and GPT-4 to auto-classify and diagnose backup failure logs, replacing manual triage for 10K+ daily incidents.
- **AI-Driven Backup Policy Recommender:** Developed an ML-based system using Python and scikit-learn to analyze historical backup patterns and recommend optimal protection policies, improving storage efficiency by 20%.
- **Native Edge Orchestrator:** Created Go-based microservices for edge node onboarding, discovery, and secure configuration management with zero-touch provisioning.
- **Object Store Cloud Backup:** Architected high-throughput upload/download modules for Azure & AWS with multi-part parallel transfers, supporting petabyte-scale distributed storage.
- **Active Directory GLR:** Built iSCSI-based live mount restore system reducing AD recovery from hours to minutes for enterprises with 100K+ directory objects.
- **Bare Metal Recovery (BMR/SSR):** Designed low-level backup/restore mechanisms in C++ and VSS enabling complete OS state recovery with data consistency guarantees across heterogeneous hardware.

PATENTS & INTELLECTUAL PROPERTY

14 US patents granted covering cloud object storage optimization, Office 365 backup strategies, Hyper-V virtualization recovery, incremental data deduplication, and distributed data protection architectures.

US 11,095,745; US 11,829,630; US 12,216,549; US 11,836,046; US 12,105,602; US 11,809,280; US 12,625,999; US 12,032,445; US 11,340,824; US 11,442,819; US 12,026,062; US 11,669,403; US 11,734,124; US 11,442,822. Public records: patents.justia.com

EDUCATION

M.Tech in Artificial Intelligence & Machine Learning, BITS Pilani (WILP) | Pursuing (2026 - Present)
B.Tech in Information Technology, Guru Ghasidas Central University, Bilaspur | CGPA 7.96 / 10 | 2012 - 2016

ACHIEVEMENTS & LEADERSHIP

- Scrum Lead for multiple product releases in Dell's Data Protection suite, coordinating 8-member cross-functional teams.
- Delivered 10+ internal tech talks on backup architecture, cloud integration, system design, and AI/ML integration patterns.
- Championed AI-first development practices within the team, integrating LLM-based tooling into code review and incident response workflows.
- Recognized for resolving critical customer escalations (P0/P1) under tight SLAs with zero recurring incidents.
- Mentored 5+ junior and mid-level engineers on system design, distributed systems debugging, and AI-augmented development.